Emily Maxwell Outland

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Interests & Inspirations

I am driven by a simple conviction: that humans and autonomous systems are better together than either is alone. My research sits at the intersection of human factors and autonomy — asking how we design machines that genuinely serve the humans who work alongside them, rather than the other way around.

Space & Exploration	Space analogs, long-duration mission human factors, human-machine teams in operational environments
Explainable Autonomy	Interpretable AI for non-expert users, transparency in autonomous decision-making, trust calibration
Broader Curiosities	Science communication, human-centered autonomy, physiological monitoring in extreme environments

Education

BS in Computer Engineering
Terre Haute, IN
ROSE-HULMAN INSTITUTE OF TECHNOLOGY — *May 2025*
Minors: Mathematics, Music

MS in Aerospace Engineering Science
Boulder, CO
UNIVERSITY OF COLORADO BOULDER — *Expected Graduation: May 2027*
Focus Area: Bioastronautics

Experience

Air Force Research Lab - Space Vehicles Directorate
SMART SCHOLAR — *2023 - Current*
- **Summer 2023:** explored explainable machine learning methods to detect anomalies in satellite telemetry data; research published
- **Summer 2024:** explored the viability of applying genetic algorithms to control system optimization; research concluded, not published
- **Summer 2025:** explored using Large Language Models (LLMs) to assist satellite operators in task identification and execution; research in progress

ARIA Systems Lab
GRADUATE RESEARCHER — *August 2025 - Current*
Research into explainable multi-agent motion planning and human mental state modeling

Publication

An Explainable Machine Learning Approach for Anomaly Detection in Satellite Telemetry Data
E MAXWELL*, S KRICHEFF*, C PLAKS*, M SIMON — *Published: 13 May 2024*
- IEEE Aerospace Conference Proceedings 2024, Presented: 7 Mar 2024
- Applying explainability methods (SHAP, LIME, and Layer-wise Relevance Propagation) to novel IF-CBLOF (Isolation Forest Clustering-Based Local Outlier Factor) model and LSTM Neural Network to detect, score, and explain anomalies in satellite telemetry data
- Created and evaluated Hybrid IF-CBLOF Model for Anomaly Scoring
- Data analysis, labeling, and pruning

* indicates equal contribution as first authors

Projects

The Mad HATter
EXPLORING THE EFFECTS OF AUTONOMOUS AGENT PERSONALITY ON HUMAN-AGENT TEAM (HAT) DYNAMICS IN A COLLABORATIVE SETTING — *Aug 2025 – Dec 2025*
- Studied how AI apology tone and error rate affect human trust, workload, and agent perception in a within-subjects human participant study
- Found apology quality significantly predicted trust; rude apologies produced *lower* trust than no apology, and higher apology frequency decreased trust independent of tone
- Results suggest good apologies can partially recover trust under high AI error, with direct implications for personality design in human-machine teaming

Explainable Sampling-Based Motion Planning

Comparing saliency and text-based explanations for RRT path understanding *Aug 2025 – Dec 2025*

- Developed two explanation types — visual heatmaps and interactive contrastive text — to help humans understand and predict robot motion planning decisions
- Evaluated explanation effectiveness on human path prediction accuracy via a counterbalanced within-subjects study (N = 24) across three environment types
- Identified limitations of standard path similarity metrics in complex environments, motivating topology-aware evaluation as future work

In-Progress

Evaluating Comprehension and Explainability for Multi-Agent Explainable Path Planning on a Grid World

Research Proposal Experimental Pilot *January 2026 - Current*

- Design an experiment to evaluate comprehension and intuitiveness of a multi-agent explainable planner which uses path segmentation
- Incorporate validated surveys for comprehension and trust with counterbalanced explanation conditions
- Run a small pilot test to write a proposal for the continuation of this research area

Longitudinal Affective State Trajectory Modeling

Master's Thesis *March 2026 - Current*

- Model human emotional state as a continuous trajectory over time using wearable physiological data, grounded in validated psychological frameworks (Russell circumplex, PANAS)
- Develop and extend a probabilistic deep learning architecture for irregularly-sampled multimodal time series, applied to a 4-month naturalistic dataset ($n = 71$)
- Validate model outputs against established anxiety measures; motivate application to adaptive human-robot teaming in high-stakes environments such as spaceflight

Research Methods & Expertise

Experimental Design	Within-/between-subjects, Latin square counterbalancing, IRB protocols, power analysis
Statistical Analysis	Repeated-measures ANOVA, Tukey HSD, ANCOVA, Shapiro-Wilk, Levene's test
Human Factors & HRI	NASA-TLX, trust measurement (Likert composites) and calibration, human-agent teaming
Programming	Python, Julia, R, C/C++, MATLAB, Java, AI-assisted programming
ML & Motion Planning	Reinforcement learning (MDP/POMDP), sampling planners, explainable planning, multi-agent planning
Autonomous Systems	Explainable AI (XAI), LLMs for operator assistance, satellite telemetry analysis, robotics
Technical Writing	Conference paper and presentation, peer review experience, LaTeX
Space Medicine	Autonomic nervous system physiology, HRV as a biomarker, psychosocial isolation in ICE environments
Languages	Native English, Elementary Japanese

Research Visions

- *How can we design AI systems for spaceflight rather than demanding adaptation from the human?*
- *Can HRV serve as a real-time, personalized feedback signal for astronaut cognitive and autonomic state during EVA, and can that signal inform adaptive autonomy?*
- *What does it mean for a motion planner to be truly explainable to a non-expert, and does better explanation actually produce better human-robot collaboration?*